

Comparability strategy and demonstration for post-approval production cell line change of a bevacizumab biosimilar IBI305

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ABSTRACT

High-producing cell line could improve the affordability and availability of biotherapeutic products. A post-approval production cell line change, low-titer CHO-K1S to high-titer CHO-K1SV GS-KO, was performed for a China marketed bevacizumab biosimilar IBI305. Currently, there is no regulatory guideline specifically addressing the requirements for comparability study of post-approval cell line change, which is generally regarded as the most complex process change for biological products. Following the quality by design principle and risk assessment, an extensive analytical characterization and three-way comparison was performed by using a panel of advanced analytical methods. Orthogonal and state-of-the-art techniques including nuclear magnetic resonance and high-resolution mass spectrometry were applied to mitigate the potential uncertainties of higher-order structures and to exclude any new sequence variants, scrambled disulfide bonds, glycan moiety and undesired process-related impurities such as host cell proteins. Nonclinical and clinical pharmacokinetics (PK) studies were conducted subsequently to further confirm the comparability. The results demonstrated that the post-change IBI305 was analytically comparable to the pre-change one and similar to the reference product in physicochemical and biological properties, as well as the degradation behaviors in accelerated stability and forced degradation studies. The comparability was further confirmed by comparable PK, pharmacodynamics, toxicological and immunogenicity profiles of nonclinical and clinical studies. The comparability strategy presented here might extend to cell line changes of other post-approval biological products, and particularly set a precedent in China for post-approval cell line change of commercialized biosimilars.

Statement of Significance: Our work represents the first report of comparability exercise for the post-approval production cell line change, following a QbD approach with tier-based quality attribute evaluation and risk assessment. The state-of-the-art and orthogonal techniques were applied and demonstrated the comparable structural and functional characterizations, which were further confirmed by nonclinical and clinical PK profiles.

KEYWORDS: cell line change; post-approval; comparability; quality by design; biologics

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INTRODUCTION

Change of CHO cell line from lower-titer to higher-titer production yields would reduce the environmental pressures, improve the efficiency and lower the biomanufacturing cost [1–4]. Therefore, cell line change has become extremely attractive for the marketed biosimilar or fast-follow therapeutic proteins in the developing and populous countries such as China because of fierce competition on cost of goods.

ICH, WHO, FDA and EMA have issued multiple guidelines to provide scientific and regulatory considerations for the post-approval process changes of biotherapeutic products [5–11]. Following the ICH Q5E and Q12, post-approval changes are categorized into major, moderate and minor changes based on the potential risk for safety and efficacy, to achieve risk-based management of post approval change in an efficient manner. The hierarchical strategy is typically recommended for assessing comparability between the pre- and post-change products in a step-wise approach from analytical (including both physicochemical and functional characterizations), nonclinical to clinical comparison studies. Analytical comparability provides the foundation for the demonstration of comparability. If quality differences are identified and the potential adverse impacts on safety and efficacy cannot be excluded, additional nonclinical and/or clinical studies would be warranted based on risk assessment. Existing regulatory guidelines have not yet provided clear comparability requirement for a post-approval cell line change. Because of the structural complexity and species specificity of biotherapeutic products, analytical and nonclinical studies alone might be considered inadequate to reflect all impacts on safety and efficacy, and therefore, clinical bridging study might be needed to mitigate additional comparability concerns [12].

In the last decade, considerable scientific and regulatory experience has been accumulated on the process development of therapeutic antibodies. Ever-improving analytical characterization techniques with higher sensitivity and accuracy allow detection of subtle quality differences of biotherapeutic products. For instance, high-sensitivity mass spectrometry can be used to identify any potential sequence variants, scrambled disulfide bonds, glycan moiety and residual host cell proteins (HCPs) at the low ppm level [13–15]. Based on extensive prior knowledge and in-depth structure–function analysis, the manufacturers can understand their own products more deeply over time, correctly predict the impact of process changes on critical quality attributes (CQAs), and establish scientifically reliable and robust control strategy to mitigate the potential risks. As a result, it is reasonable to assume that post-approval production cell line change following the quality by design (QbD) principle possesses relatively lower uncertainty, and the scope and extent of comparability study might be less required than similarity assessment of biosimilars, which also undergo production cell line change compared with the reference product. Therefore, for the comparability study of post-approval cell line change, a thorough analytical comparability evaluation on product quality attributes with a panel of orthogonal and complementary techniques could detect the differences between pre-change and post-change

products, and might adequately demonstrate product comparability in combination with nonclinical and clinical pharmacokinetics (PK)/pharmacodynamics (PD) bridging studies. A clinical efficacy trial would be conducted if any CQA related to efficacy cannot be demonstrated comparable.

IBI305 (BYVASDA®), a bevacizumab biosimilar developed by Innovent Biologics, was initially approved by the National Medical Products Administration (NMPA) of China in 2020. IBI305 is a recombinant humanized anti-VEGF monoclonal antibody, which binds VEGF selectively with high affinity and blocks VEGF binding to its receptors on the surface of vascular endothelial cells. IBI305 had undergone a major post-approval change shortly after marketing in 2020 to increase the manufacturing scale from 1000 L scale to 3000 L scale, and the scale change was approved in 2021. Both the initial development and post-approval scale-up change followed the QbD principle to evaluate CQAs and the potential impact of the process change on CQAs.

To further improve the availability and reduce the manufacturing cost of IBI305, a post-approval cell line change with corresponding process adjustments was carried out here, with expression titer significantly increased by approximately three folds from the host cell line CHO-K1S to CHOK1SV GS-KO. To demonstrate the comparability for this cell line change, based on prior knowledge and accumulative experience in process change and a risk assessment of the potential impacts on CQAs, a comprehensive and three-way analytical comparability study was performed among pre-change IBI305, post-change IBI305 and the reference product Avastin®, followed by confirmatory non-clinical and clinical PK and safety studies. The analytical results indicated that the post-change IBI305 was highly comparable to the pre-change one, and substantially similar to the reference product, with risks further mitigated by subsequent nonclinical and clinical PK and safety studies.

MATERIALS AND METHODS

Materials

Twenty-two lots of reference product Avastin® (100 mg/4 mL, China-registered) were purchased from Roche over a period of ~5 years. Eighteen lots of pre-change IBI305 and four lots of post-change IBI305 products (100 mg/4 mL) were manufactured by Innovent Biologics (Suzhou) Co., Ltd in China. The detailed information of samples is listed in Table S1.

Methods

The intact and reduced molecular weights, non-reduced and reduced peptide mapping, glycan mapping, free sulfhydryl, isoelectric point, size variants, charge variants, higher-order structures (HOS) by circular dichroism (CD) and differential scanning calorimetry (DSC), subvisible particles, VEGF-binding activity, Fc receptors and C1q-binding affinities, process-related impurities were analyzed as previously described [16].

HOS analysis by nuclear magnetic resonance

Nuclear magnetic resonance (NMR) data were acquired at 310 K on a Bruker Avance III 900 MHz spectrometer equipped with a 5 mm-CPTCI cryogenically cooled probe and a Z-axis gradient system. The one-dimensional (1D) ^1H NMR and two-dimensional (2D) ^1H - ^{13}C NMR profile spectra were collected using a Bruker standard experiment of zgpg30 and hsqcgp30, respectively [17, 18]. The acquisition time was 1.5 s and relaxation delay was 3.0 s. The free induction decay accumulations consisted of 40 960 complex points. The total acquisition time was 1 h for 1D ^1H NMR and 4 h for 2D ^1H - ^{13}C NMR. All spectral processing was performed with Topspin 3.5 software.

Residual process-related impurities assays

The host cell DNA (HCD) was detected by utilizing real-time fluorescence quantitative polymerase chain reaction (qPCR) technology on Thermo Fast 7500 qPCR instrument. A commercially available enzyme linked immunosorbent assay (ELISA) kit (ADI-900-057, Enzo) was used to quantify residual Protein A.

The residual HCPs in pre- and post-change process intermediates and drug substances were characterized and compared using orthogonal methods, including HCP ELISA, sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) and nanoLC-MS/MS. A commercial CHO HCP ELISA kit (Cygnus Technologies) was used to quantitatively measure the total HCP amount for intermediate quality control and lot release. The silver-stained 1D-gel electropherograms was conducted to support the comparison results measured by HCP ELISA. To further detect and identify individual HCPs before and after the cell line change, nanoLC-MS/MS technique was performed on the Easy-nLC 1200 system coupled to the Q Exactive HF-X Orbitrap mass spectrometer (Thermo Fisher Scientific). Offline pH fractionation using the Pierce high pH reversed-phase peptide fractionation kit (Thermo Fisher Scientific) was conducted to improve detection sensitivity. The MS/MS data acquired data-dependently were searched against a customized protein database composed of sequences obtained from the CHO fasta database using the MaxQuant (Max Planck Institute of Biochemistry, Germany). The peptide identification was based on a false discovery rate below 0.01. A minimum of two unique peptides over all data sets were required for identification.

Accelerated stability and forced degradation study

Accelerated stability comparability was assessed using pre- and post-change IBI305 under accelerated condition at $25^\circ\text{C} \pm 2^\circ\text{C}$ for 6 months. Three lots of pre-change IBI305, post-change IBI305 and Avastin were selected for forced degradation study. The degradation profiles of pre- and post-change IBI305 and Avastin[®] were investigated side-by-side under the stressed conditions of high temperature (40°C) and light exposure (5000 ± 500 lux) for 0, 3, 5 and 10 days. Subsequent evaluation of stability-indicating attributes for the degradation samples was

carried out using liquid chromatography-mass spectrometry (LC-MS), size exclusion liquid chromatography (SEC-HPLC), non-reduced and reduced capillary electrophoresis sodium dodecyl sulfate (CE-SDS), cation exchange chromatography (CEX-HPLC) and potency assay. Other general properties, such as appearance, pH and concentration, were also tested. The conditions, sampling time points and inspection items for the accelerated stability and forced degradation studies are shown in Table S2.

Nonclinical study

In vitro PD comparability studies were performed using a luciferase reporter cell assay for blocking activity and primary human umbilical vein endothelial cells (HUVEC) for inhibition of cell proliferation. For *in vivo* PD comparability study, HL60 tumor-bearing NOD-SCID mice were used to investigate *in vivo* anti-tumor efficacy, and the IgG1-treated group was set as the control group. The PK comparability study was conducted in BALB/c mice and cynomolgus monkeys following a single intravenous dose, accompanied by immunogenicity testing in monkeys. And the Avastin[®]-treated group was set as the reference in the above studies. In mice study, 27 female mice were randomly divided into pre-change IBI305 group, post-change IBI305 group and Avastin[®] group. Mice were intravenously administered with a single AT 10 mg/kg. Blood samples were collected at various post-dose time points to determine the serum concentration. In cynomolgus monkey study, 18 cynomolgus monkeys were randomly divided into three groups with three male and three female monkeys in each, including pre-change IBI305 group, post-change IBI305 group and Avastin[®] group. Monkeys were intravenously injected with a single dose of 15 mg/kg. Blood samples were collected at various time points for concentration determination and antidrug antibodies (ADA) analysis. The comparability among pre- and post-change IBI305 and Avastin[®] was evaluated based on the PK parameters, including C_{max} , AUC_{last} and AUC_{inf} .

A 4-week repeat-dose toxicity study in cynomolgus monkeys was performed to assess toxicity profile. In this study, 40 cynomolgus monkeys were randomly assigned to four groups with five male and five female monkeys in each, including control group, pre-change IBI305 group, post-change IBI305 group and Avastin[®] group. Monkeys were intravenously administered with IBI305 buffer or test articles at 50 mg/kg once a week for 4 consecutive weeks (totally five doses) followed by a 4-week recovery phase. The comparability of safety profile among pre- and post-change IBI305 and Avastin[®] was evaluated based on toxicity profile comparison.

Clinical PK study

A randomized, double-blind, two-arm, parallel, controlled, single-dose phase I study was designed to further assess the comparability of pre- and post-change IBI305 in healthy male subjects (chinadrugtrials.org CTR20211576). According to the recommendation for volunteer selection in *Guidelines for Clinical Trials in Biosimilars of Bevacizumab*

Injection issued by NMPA CDE, male health subjects were selected for the clinical PK study. A total of 100 healthy male subjects were randomized in a 1:1 ratio to receive a single-dose intravenous infusion of 3 mg/kg pre- and post-change IBI305. PK and immunogenicity at various time points after drug administration were analyzed. The primary endpoints were area under the concentration-time curve from 0 to infinity ($AUC_{0-\infty}$). The secondary endpoints included the other PK parameters, immunogenicity and safety measurements.

Blood samples were collected at the following time points: before infusion, immediately and various time-points after infusion. PK parameters were estimated using non-compartmental analysis (Pkanalix 2021R1, Lixoft, Antony, France). The criteria for the PK comparability between pre- and post-change IBI305 were 0.80–1.25 for the 90% confidence interval of test-to-reference ratios of $AUC_{0-\infty}$.

Subjects received physical examinations, vital sign assessment, electrocardiography, blood chemistry and complete blood count tests during the study. Abnormal results were reviewed by an investigator. Clinically significant abnormal laboratory results relative to those at baseline and the screening stage were recorded as adverse events. AEs of special interest in this study included high blood pressure, gastrointestinal perforation, proteinuria, phlebitis, bleeding events, cardiotoxicity, thrombosis, fistula, reversible posterior encephalopathy syndrome, hypersensitivity and infusion reaction. The severity of an AE was assessed based on Common Terminology Criteria for Adverse Events version 5.0. AEs were coded in accordance with the Medical Dictionary for Regulatory Activities (v24.0). Blood samples were collected for immunogenicity testing (ADA/neutralizing antibodies).

RESULTS

Summaries on the changes and the comparability exercise

A post-approval production cell line change was conducted for IBI305 from the cell line CHO-K1S to CHOK1SV GS-KO, both of which locate closely in the CHO cell lineage [19, 20]. The upstream and downstream processes were optimized accordingly to achieve higher-titer production, including the corresponding changes of some raw materials, process steps and operation parameters, as summarized in Table 1. Based on the understanding of process and product, as well as the historical data and pre-defined CQAs, a risk assessment was conducted to evaluate the impact of each change on the quality, safety and efficacy of IBI305 (Table 1). The risk assessment facilitated the application of hierarchical strategy for comparability exercise, and setting up accordingly a comprehensive comparability protocol, integrating in-process control, analytical, nonclinical and clinical PK and safety comparison studies, to demonstrate the comparability between the pre- and post-change IBI305.

In terms of analytical comparability study, a broad panel of highly sensitive, orthogonal and state-of-the-art analytical techniques was applied to characterize IBI305 quality attributes, and special attention was paid to the quality

attributes susceptible to be impacted as assessed in Table 1, such as sequence variants, HOS and HCP profile. Comparability acceptance criteria were established by referring to the similarity acceptance criteria submitted in the new drug application (NDA) in China [16], which were established based on the three-tier-based quality attribute evaluation [21], as shown in Table 2. Regarding to the samples (Table S1), all 18 lots of pre-change IBI305 drug products for clinical studies and process performance qualification (PPQ) and four lots of post-change drug products, including one engineering lot and three PPQ lots, were involved in the comparability exercise. A total of 22 lots of Avastin[®] sourced over ~5 years were tested for Tier 1 and 2 attributes, and the results were subjected to statistical calculation of the quality range (mean $\pm$ 3 SD) as the similarity acceptance criteria. The representative lots of pre-change IBI305, post-change IBI305 and Avastin[®] (3:3:3) were selected for a side-by-side comparison of Tier 3 attributes. The quality attributes, techniques, tiers, comparability acceptance criteria and results are shown in Table 2. The suitability of existing analytical procedures was also evaluated using the new process material, and each analytical method was either validated or qualified. The comprehensive analytical results demonstrated that the post-change IBI305 was comparable to the pre-change one and similar to Avastin[®], which was further confirmed by non-clinical and clinical PK studies.

Process comparison

The new host cell line CHOK1SV GS-KO was developed from CHO-K1S cell line through glutamine synthetase knockout for achieving significant productivity improvements in bulk cell cultures [22]. A comparison of the process performance was conducted for the pre- and post-change processes at 3000 L scale. Better cell growth performances with higher viable cell densities and cell viabilities were observed after cell line change (Fig. 1). Almost 4-fold higher level of expression titers was achieved. CQAs of in-process intermediates before and after the change were compared, and the results indicated that the post-change process possessed equivalent capabilities for impurity removal compared with the pre-change process (data not shown). In particular, the presence and removal of HCPs during the critical downstream steps were compared between the pre- and post-change processes (see the Process-Related Impurities section).

Primary structure

The primary structure was investigated by multiple advanced and complementary methods, including intact and reduced LC-MS molecular mass analysis, reduced and non-reduced liquid chromatography tandem mass spectrometry (LC-MS/MS) peptide mapping, glycan mapping and free sulfhydryl quantification. The results demonstrated that the primary structures of the pre- and post-change IBI305 were highly comparable and similar to those of Avastin[®].

Cell line change could potentially introduce unintended amino acid substitutions, which raised special attention in

Table 1. Details of the cell line change, relevant process adaptations and the corresponding risk assessment

Category		Pre-change	Post-change	Rationale	Potentially impacted quality attribute
Cell bank	Cell line	CHO-K1S	CHOK1 SV GS-KO	Increase titer	Sequence variants, purity and product-related impurities, charge variants, HCP, HOS
	Expression vector	pKN006/pKN001	pXC-17.4/pXC-18.4	Match the cell line	
Raw materials	Media	Commercial media	In-house media	Increase titer and decrease cost	Sequence variants, purity and product-related impurities, charge variants, HCP, HOS
	Resin	Mabselect SuRe	Mabselect PrismA	Meet higher amounts	Purity and product-related impurities, charge variants, HCP, HCD, Protein A
		Fractogel EMD SO3-(M)	Capto S ImpAct	Meet higher amounts	Purity and product-related impurities, charge variants, HCP
	Ultrafiltration membrane	Pellicon® 2	Pellicon® 3	Meet higher concentration of DS	Aggregates
DS Process	Upstream-cell culture	Adjust parameters of seeding density, temperature and pH		Increase titer	Sequence variants, purity and product-related impurities, charge variants, HCP, HOS
	Downstream-purification	Use Triton X-100 to replace CA (caprylic acid) in the S/D virus inactivation step Remove HIC step		Triton X-100 easier to remove Useless in virus clearance and purity improvement	Virus clearance Purity and product-related impurities, HCP, virus clearance
DS specification	Protein content	30 mg/ml	60 mg/ml	Decrease storage packages	Stability
DP process	lot size	35 000 vials	45 000 vials	Meet higher amounts	Stability
	Excipients	Vendor of sodium acetate and acetic acid changed to Merck KGaA		Meet both USP and ChP	Stability, safety

Table 2. Summarized quality attributes, analytical methods and results in comparability assessment among the pre- and post-change IBI305 and Avastin®

Category	Attributes	Methods	Risk	Tier	Similarity Acceptance Criteria	Results/min-max range (n)	
						Post-change IBI305	Pre-change IBI305
Primary structure	Amino acid sequence	LC-MS/MS	High	3	Identical to the RP	Identical to the RP (3:3:3)	
	Intact and reduced molecular weights (Da)	LC-MS	Moderate	3	±50 ppm	Similar to the theoretical mass (3:3:3)	
	Glycosylation site	LC-MS/MS	Low	3	Identical to the RP	HC Asn303	HC Asn303
	Glycan profile	HILIC-FLD	Moderate/low	2/3	88.1–97.1	92.4–92.6 (4)	92.9–94.2 (18)
					Report data	12.2–15.2 (4)	14.0–24.5 (18)
					Report data	4.0–4.4 (4)	1.7–3.0 (18)
					≤1.2	0.7–0.8 (4)	0.9–1.2 (18)
	Disulfide linkage	LC-MS/MS	Moderate	3	Identical to the RP	Identical to the RP (3:3:3)	
	Free thiols (mol SH/mol IgG)	Ellman's assay	Moderate/low	3	Report data	0.27–0.31 (3)	0.14–0.17 (3)
	PTM and sequence variants (%)	LC-MS/MS	Moderate	3	Report data	2.2–2.3 (3)	2.2–2.3 (3)
HOS	Oxidation	HC Asn390/395	Moderate	3		0.1–0.2 (3)	0.1–0.2 (3)
						1.5–1.6 (3)	0.4–0.7 (3)
						4.1–5.1 (3)	3.7–7.2 (3)
						1.0–2.6 (3)	0.9–1.0 (3)
	Glycation	LC K188	Moderate	3		0.4–0.4 (3)	0.5–0.5 (3)
						1.2–1.3 (3)	1.1–1.1 (3)
							1.8–1.8 (3)
	HC N-terminal pyro-Glu		Moderate	3			0.18–0.18 (3)
							2.6–2.8 (3)
Charge variants	Secondary and tertiary structure	Far/Near-UV CD 1D ¹ H NMR 2D ¹ H- ¹³ C NMR DSC	Moderate	3	Spectra comparable	Comparable	Comparable
						Comparable	Comparable
						Comparable	Comparable
						72.0–72.1 (3)	72.1–72.2 (3)
Size variants	Thermal stability	CEX-HPLC	Moderate	2	Thermograms comparable	84.4–84.6 (3)	84.3–84.6 (3)
						23.7–25.5 (4)	20.1–22.7 (18)
						65.0–72.4	68.3–71.5 (18)
						9.5–10.2 (4)	8.2–9.9 (18)
Charge variants	T _{m1} (°C) T _{m2} (°C)	iCIEF	Moderate	3		8.4–8.4 (3)	8.4–8.4 (3)
	Acidic species (%)	SEC-HPLC	Moderate	2		0.6–0.7 (4)	0.3–0.6 (18)
						99.3–99.4 (4)	99.4–99.7 (18)
						155.3–155.8 (3)	155.7–156.2 (3)
	Basic species (%)	SEC-MALS	Moderate	3	Chromatograms comparable, report data		
Size variants	Aggregates (%)	MW (kDa)	Moderate	2			
	Monomer (%)	MW (kDa)	Moderate	3			
	Aggregates	MW (kDa)	Moderate	2			
Size variants	Intact IgG (%)	nrCE-SDS	Moderate	2			
	HC and LC content (%)	rCE-SDS	Moderate	2			
	NGHC (%)	rCE-SDS	Moderate	2			

(continued)

Table 2. Continued

Category	Attributes	Methods	Risk	Tier	Similarity Acceptance Criteria	Results/min-max range (n)	Pre-change IBI305	Pre-change IBI305	Avastin®
Process-related impurities	HCP	ELISA (ng/mg)	Moderate	2	Report data	<15 (4)	<15 (18)		N/A
		SDS-PAGE	Moderate	3	Report data	Comparable	Comparable		N/A
	HCD (ng/dose) Protein-A (ng/mg)	LC-MS/MS	Moderate	3	Report data	Comparable	Comparable		N/A
		qPCR	Moderate	2	Report data	<0.2 (4)	<0.2 (20)		N/A
		ELISA	Moderate	2	Report data	<0.08 (4)	<0.02 (20)		N/A
Content	Concentration (mg/ml)	UV Method	High	1	22.5–27.5	24.8–25.0 (4)	24.8–25.2 (18)		N/A
Bioactivity	Potency (%)	Cell-based assay	High	1	75–125	88–97 (4)	90–113 (18)		79–123 (17)
	VEGF-A-binding activity (%)	ELISA	High	3	75–125	83–102 (3)	89–102 (3)		102–113 (3)
Immunological properties	FeRn binding (KD, M)	BLI	Moderate	3	Comparable on the order of magnitude	2.7×10^{-8} – 2.7×10^{-8} (3)	2.6×10^{-8} – 2.7×10^{-8} (3)	2.6×10^{-8} – 2.7×10^{-8} (3)	2.6×10^{-8} – 2.7×10^{-8} (3)
	FcγRI binding (KD, M)		Low	3		2.0×10^{-9} – 2.5×10^{-9} (3)	1.9×10^{-9} – 2.3×10^{-9} (3)	1.6×10^{-9} – 1.8×10^{-9} (3)	1.6×10^{-9} – 1.8×10^{-9} (3)
	FcγRIIa binding (KD, M)		Low	3		5.0×10^{-7} – 5.6×10^{-7} (3)	5.2×10^{-7} – 5.5×10^{-7} (3)	9.6×10^{-7} – 14.3×10^{-7} (3)	9.6×10^{-7} – 14.3×10^{-7} (3)
	FcγRIIb binding (KD, M)		Low	3		2.2×10^{-7} – 2.5×10^{-7} (3)	2.2×10^{-7} – 2.9×10^{-7} (3)	1.5×10^{-7} – 2.1×10^{-7} (3)	1.5×10^{-7} – 2.1×10^{-7} (3)
	FcγRIIIa (F158)-binding (KD, M)		Low	3		1.5×10^{-7} – 1.7×10^{-7} (3)	2.5×10^{-7} – 3.3×10^{-7} (3)	1.6×10^{-7} – 2.1×10^{-7} (3)	1.6×10^{-7} – 2.1×10^{-7} (3)
	FcγRIIIa (V158) binding (KD, M)		Low	3		1.1×10^{-7} – 1.2×10^{-7} (3)	0.8×10^{-7} – 1.4×10^{-7} (3)	1.1×10^{-7} – 2.3×10^{-7} (3)	1.1×10^{-7} – 2.3×10^{-7} (3)
	FcγRIIIb binding (KD, M)		Low	3		2.1×10^{-6} – 3.1×10^{-6} (3)	4.6×10^{-6} – 7.3×10^{-6} (3)	2.0×10^{-6} – 2.7×10^{-6} (3)	2.0×10^{-6} – 2.7×10^{-6} (3)
	C1q binding (KD, M)		Low	3		1.3×10^{-7} – 1.3×10^{-7} (3)	1.3×10^{-7} – 1.3×10^{-7} (3)	1.5×10^{-7} – 1.6×10^{-7} (3)	1.5×10^{-7} – 1.6×10^{-7} (3)
	Subvisible particles	HIAC	Moderate	2	Report data	36–113 (4)	24–329 (18)		N/A
		(particles/vial)				0–2 (10)	0–6 (10)		N/A
Particles	MFI	(particles/ml)	Moderate	3	Report data	9942–10,484 (3)	5386–10,641 (3)	1772–2549 (3)	
						529–664 (3)	428–619 (3)	349–548 (3)	
						44–50 (3)	6–78 (3)	154–245 (3)	
						4–14 (3)	0–44 (3)	38–65 (3)	
Forced degradation	High temperature (40°C ± 2°C) Illumination (5000 ± 500 lx)	Purity and binding activity assays	Moderate	3	Similar trend	Similar trend	Similar trend	Similar trend	Similar trend
Stability	Long-term stability (5°C ± 3°C)	Stability assays	Moderate	3	Similar trend	Similar trend	Similar trend	Similar trend	N/A
	Accelerated stability (25°C ± 2°C)								
	Stressed stability (40°C ± 2°C)								

^aThe results of process-related impurities derived from the assays results of corresponding drug substances. RP means reference product. N/A means not applicable

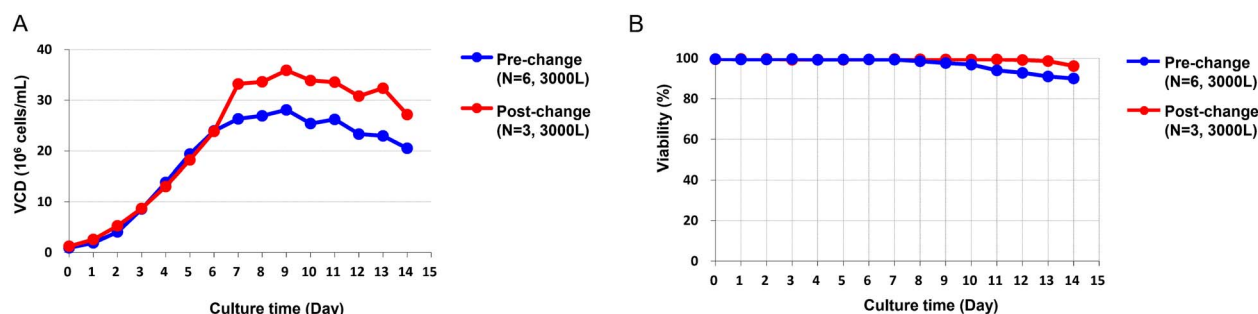


Figure 1. Cell growth results of the pre- and post-change 3000 L production bioreactor. (A) Viable cell density. (B) Cell viability.

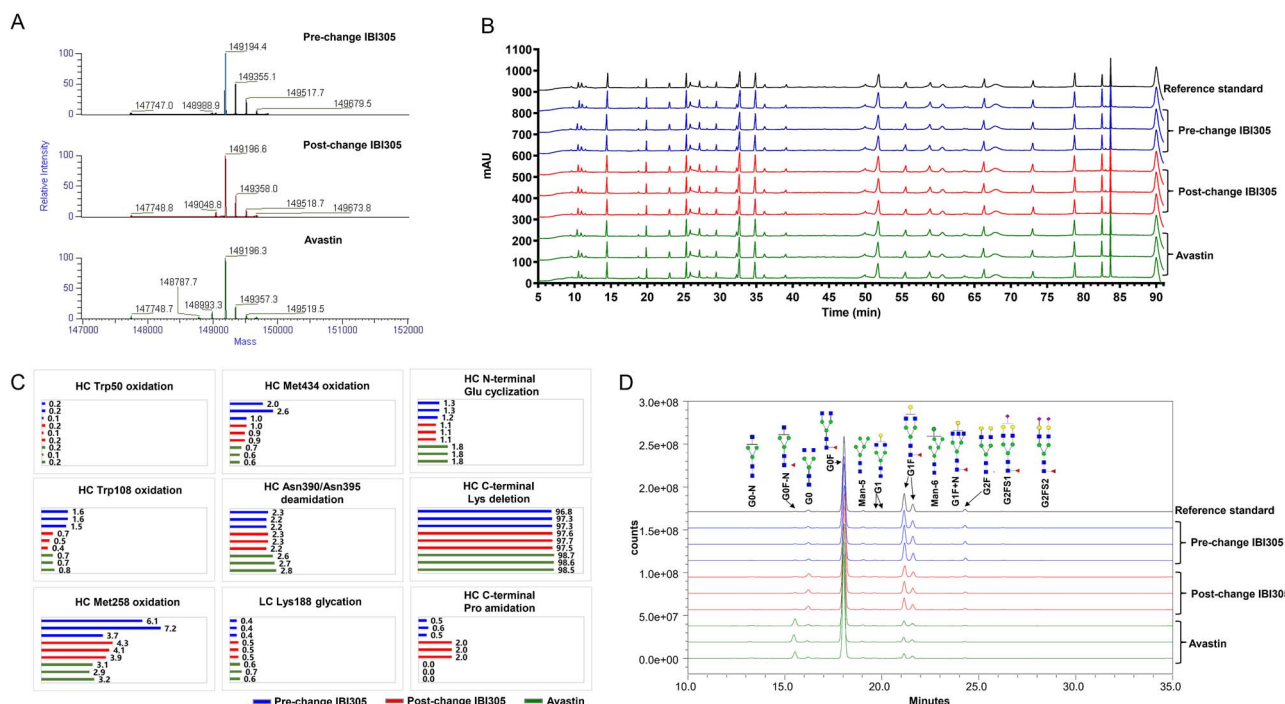


Figure 2. Primary structure of the pre-change, post-change IBI305 and Avastin[®]. (A) The mass spectra deconvolution of intact molecular mass, (B) Reduced peptide mapping of Lys-C digestion, (C) PTMs and (D) N-glycan profiles by HILIC-FLD.

the amino acid sequence analysis. 0.1% at an individual amino acid site is proposed as an appropriate sequence variant control limit for biologics development [23, 24]. The intact, deglycosylated and reduced molecular masses of the pre- and post-change IBI305 and Avastin[®] determined by LC-MS closely matched the theoretical masses of Avastin[®] (Fig. 2A), and no additional mass species was detected from the post-change IBI305 alone. The reduced peptide maps of the pre- and post-change IBI305 and Avastin[®] showed comparable LC-UV/MS peak profiles, with comparative retention times and intensities, and no additional new peaks were observed in post-change IBI305 (Fig. 2B). No obvious sequence variant ($\geq 0.1\%$) was detectable in the tested samples.

The similar sites and levels of posttranslational modifications (PTMs) were observed among all samples by reduced LC-MS/MS peptide mapping (Fig. 2C), except for slightly higher levels of Met258 and Met434 oxidation in two pre-change IBI305 lots because of the relatively longer

storage time at the recommended storage condition. The abundances of C-terminal proline amidation in the post-change IBI305 were slightly higher than those in the pre-change IBI305 and Avastin[®], which may contribute to the higher levels of basic species in the post-change IBI305 lots [25]. Slightly higher levels of C-terminal lysine truncation and N-terminal glutamic acid cyclization were observed in post-change IBI305, which are acknowledged to have no adverse impact on the efficacy and safety [26–28].

The overlay of glycan maps by hydrophilic interaction liquid chromatography with fluorescence detection (HILIC-FLD) is shown in Fig. 2D. Similar glycan types and comparable intensities were observed for all products, with no new peaks for the post-change IBI305. As shown in Fig. S1, the levels of main fucosylated glycoforms (G0F + G1F + G2F) were comparable among all products and within comparability acceptance criteria. The content of Man-5 was slightly lower after the cell line change but more similar to Avastin[®]. Slightly higher proportions of

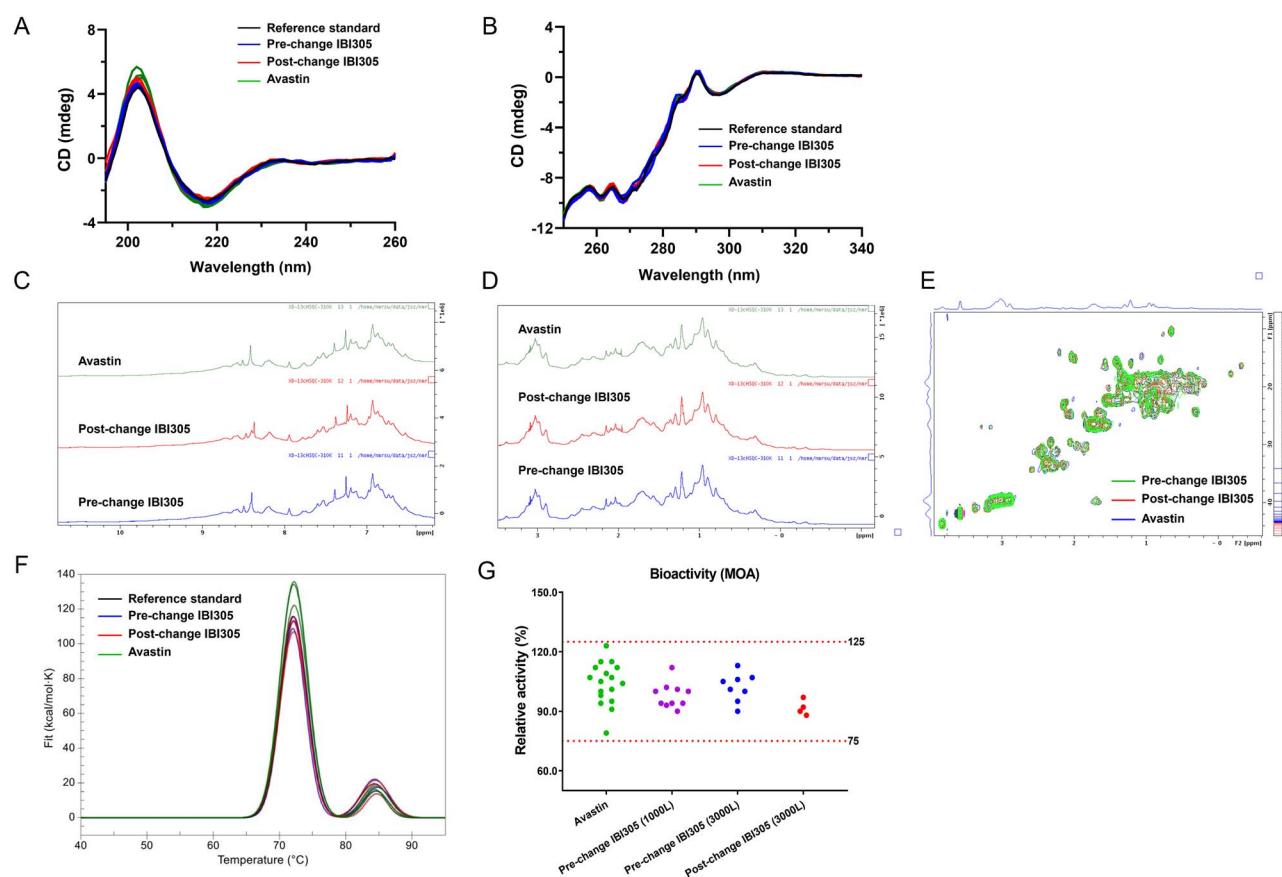


Figure 3. Comparison of HOS and bioactivity of the pre-change, post-change IBI305 and Avastin[®]. (A) Far-UV CD, (B) near-UV CD, (C) amino NMR spectrum, (D) methyl NMR spectrum, (E) 2D ¹H-¹³C NMR spectrum, (F) DSC thermogram and (G) bioactivity.

afucosylation (G0-N + G0 + G1 + G2) and lower levels of galactosylation (G1F + G2F) were observed after the change but more similar to Avastin[®].

Non-reduced LC-MS/MS peptide mapping analysis indicated identical disulfide linkages for all products, without any detectable disulfide scrambling (Fig. S2). Increasing evidence has related protein stability, aggregation and affinity to free sulfhydryl, which is most likely because of incomplete formation of disulfide bonds [29–31]. The content of free sulfhydryl measured by Ellman's assay was slightly higher after the change but still at acceptable low levels (0.27–0.31 mol/mol). LC-MS/MS results indicated that the increased free sulfhydryl groups mainly distributed on the cysteine residues expected to form intra-chain disulfide bonds, particularly located in the C_H2 and C_H3 domains. Such minor difference in free sulfhydryl was suspected to have no impact on disulfide linkage pattern and biological activities, which was confirmed by comparable results of HOS and functions.

Higher-order structure

The HOS of protein is a key CQA directly related to the structural integrity, stability, safety and efficacy of therapeutic proteins [32, 33]. Thorough characterization and comparison of HOS was performed by multiple orthogonal

biophysical methods, including far- and near-UV CD, 1D NMR, 2D NMR and DSC.

No difference in the far- and near-UV CD spectra was observed (Fig. 3A and B). Furthermore, both 1D NMR and 2D NMR spectra of post-change IBI305 were visually consistent with those before the change and highly similar to those of Avastin[®] (Fig. 3C–E). Overlays of 2D NMR spectra (Figs 3E and S3) showed highly similar chemical shifts of individual peaks, with consistent width and relative intensity. The pairwise comparison demonstrated that the similarity degrees were more than 95% between Avastin[®] and IBI305 (for both pre- and post-change, Fig. S3A and B) and ≥98% between the pre- and post-change IBI305 (Fig. S3C), indicating the highly consistent tertiary structure. The DSC thermograms of post-change IBI305 showed comparable thermal unfolding profiles and thermal transition temperatures (*T_m*) to the pre-change IBI305 and similar to Avastin[®] (Fig. 3F). The *T_{m1}* and *T_{m2}* of pre- and post-change IBI305 were within ±1°C of the corresponding average temperatures of Avastin[®], indicating comparable thermodynamic properties.

Collectively, CD, NMR and DSC results demonstrated that the HOS of post-change IBI305 were highly comparable to those of pre-change IBI305 and similar to those of Avastin[®]. In addition, the highly comparable biological

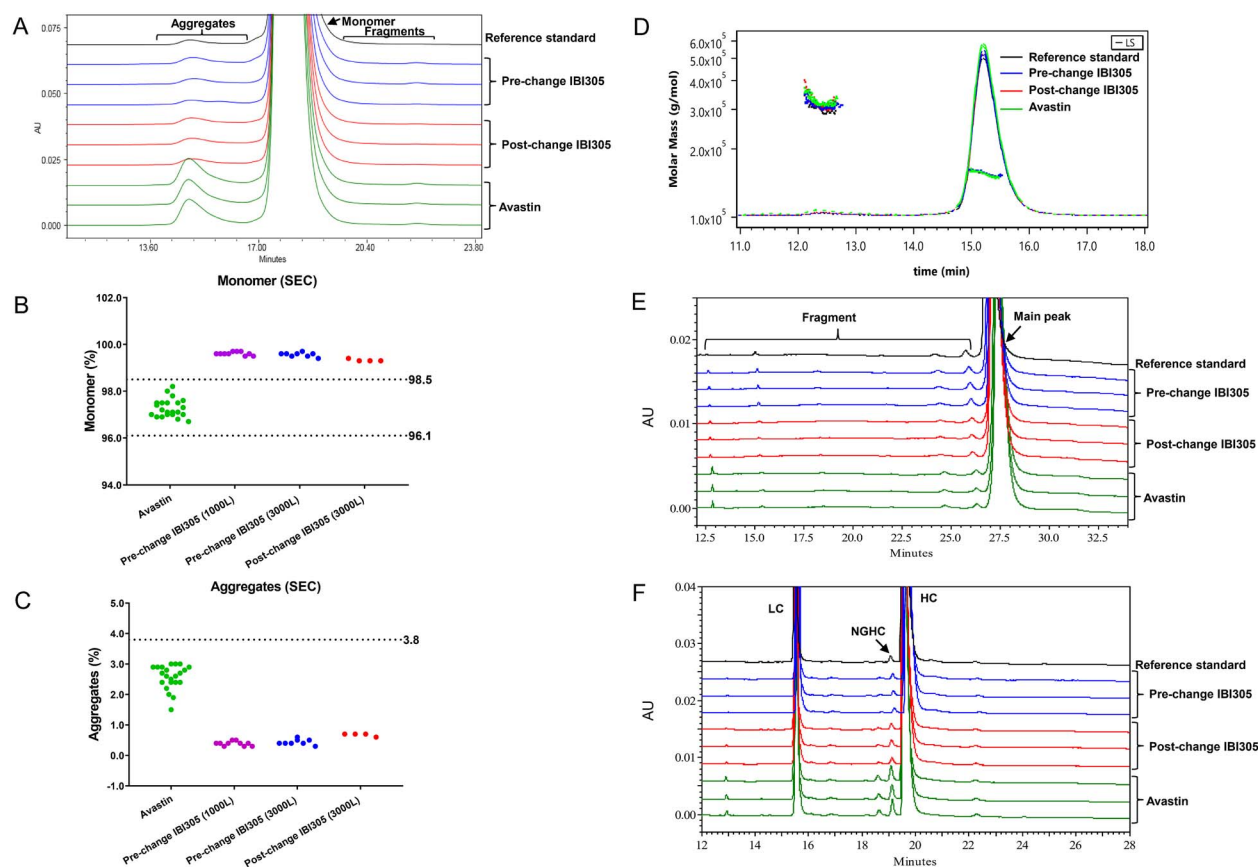


Figure 4. Comparison of size variants of the pre-change, post-change IBI305 and Avastin[®]. (A) SEC-HPLC profile, (B) scatter plot of monomer, (C) scatter plot of aggregate, (D) SEC-MALS profile, (E) non-reduced CE-SDS electrophoretogram and (F) reduced CE-SDS electrophoretogram.

activity (Fig. 3G) described below also indicated comparable conformational structure.

Purity and size variants

Aggregates were determined by SEC-HPLC and characterized by size-exclusion chromatography coupled to multi-angle light scattering (SEC-MALS). The SEC-HPLC profiles were comparable among all samples (Fig. 4A). The purity levels of monomer for pre- and post-change IBI305 were higher than those of Avastin[®] and beyond the acceptance criteria (Fig. 4B), and corresponding lower levels of aggregates were observed in IBI305 (Fig. 4C). The aggregates were characterized as dimer species by SEC-MALS for all samples (Fig. 4D). Such differences in size variants were commonly reported for marketed Avastin[®] biosimilars. The purity differences were presumed to be mainly attributed to the ages of Avastin[®] lots and have been previously demonstrated to be clinically meaningless [34–36].

Denatured size variants were assessed by non-reduced and reduced CE-SDS, and the electropherograms of the post-change IBI305 were similar to those of pre-change IBI305 and Avastin[®] without any additional new peak (Fig. 4E–F). The intact IgG purity of post-change IBI305 was comparable to that of the pre-change IBI305 and slightly higher than that observed in Avastin[®]. The types of fragment species were similar and mainly related to the

reduction of inter-chain disulfide bonds such as HHL, HL and L, with a total content lower than 2.0%. The reduced CE-SDS analysis showed that the HC + LC contents of pre- and post-change IBI305 were comparable and marginally higher than that of Avastin[®], with the contents of non-glycosylated heavy chain (NGHC) below 0.9% (Table 2).

Charge variants

The pre- and post-change IBI305 and Avastin[®] exhibited similar peak distribution with charged isoforms resolved to seven peaks detected by CEX-HPLC (Fig. 5A). As shown in the scatter plots (Fig. 5B–D), slightly lower levels of main peak and corresponding higher proportions of acidic species and basic species were observed for the post-change IBI305, while still met the comparability acceptance criteria. Actually, the acidic species contents of post-change IBI305 were more similar to those of Avastin[®] (Fig. 5B).

To further characterize the charge variants, seven charge isoforms were fractionated and analyzed by a set of physicochemical and functional assays. The collected fractions were reinjected to CEX-HPLC for confirmation and purity analysis, and the results of post-change IBI305 were shown in Fig. S4. The PTMs results of fractionated charge variants for post-change IBI305 were shown in Table S3. The characterization results indicated that the structural

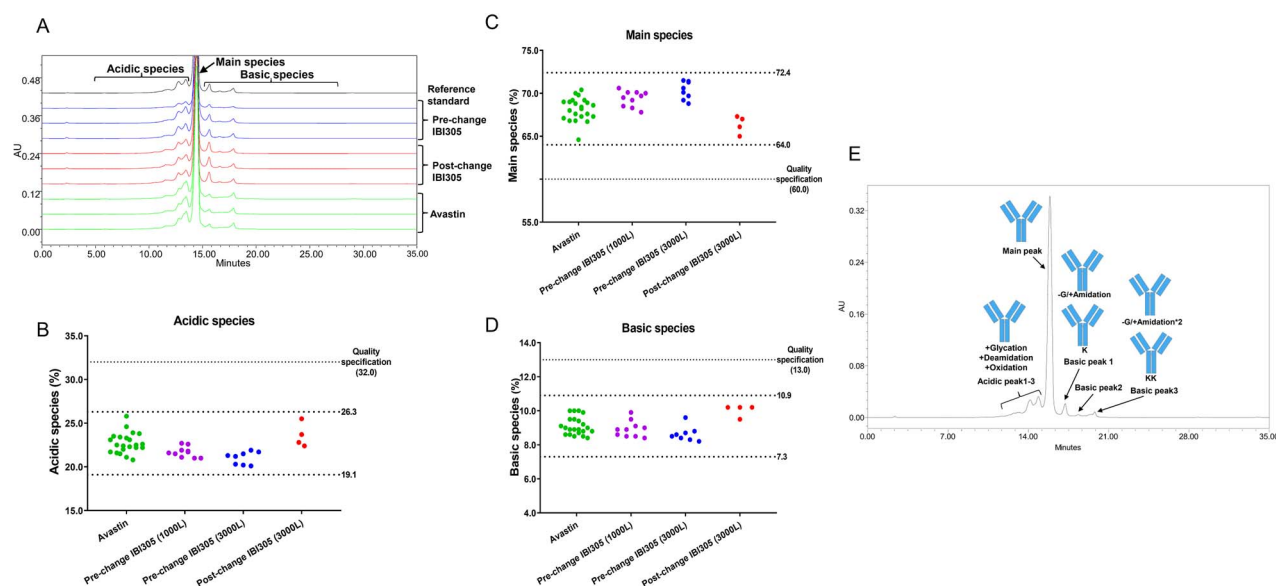


Figure 5. Charge variant profiles of the pre-change, post-change IBI305 and Avastin[®]. (A) CEX chromatogram, (B–D) scatter plot of acidic, main and basic species and (E) characterization results of acidic, main and basic species.

compositions of each charge isoform were the same among the pre-change and post-change IBI305 and Avastin[®], shown in Fig. 5E. Molecular mass results showed that the basic species were mainly attributed to the C-terminal variants with remaining lysine or proline amidation, which were consistent with the results of PTMs described above. These C-terminal variants are not expected to affect biological activities [26–28]. Interestingly, the majority of collected basic peak 2 fraction turned into the main peak when reinjected to CEX-HPLC because of dissociable aggregates (Fig. S4), and the phenomenon was also observed for both the pre-change IBI305 and Avastin[®]. The results of molecular mass and PTMs were consistent with the phenomenon observed in CEX-HPLC. The molecular mass results of basic peak 2 were the same as those of the main peak, and the sites and contents of PTMs for basic peak 2 were consistent with those of the main peak, except for slightly higher levels of remaining C-terminal lysine and C-terminal proline amidation. A further analysis by CEX-MALS revealed that the basic peak 2 fraction contained increased proportions of aggregates, which might transfer back to monomer (the main peak). The acidic species possessed relatively higher levels of deamidation, glycation and oxidation at the modification sites characterized above. Besides, all fractionated charge variants exhibited comparable biological activity.

Process-related impurities

Process-related impurities, such as HCD, HCP and residual protein A, have potential adverse impacts upon safety and are under tight control by release assays. The detection results from corresponding drug substance batches indicated that these process-related impurities were at substantially low levels (below the limit of quantification (LOQ)) for both IBI305 products before and after the change.

Cell line changes might raise more regulatory concerns about the safety issues because of the potentially altered HCD and HCP profiles. The new host cell line CHOK1SV GS-KO was derived from CHO-K1S through glutamine synthetase knockout and both cell lines shared almost the same genome sequence, thus the HCD residue would not be a concern and the detection results confirmed comparable HCD before and after the change (Table 2). According to the risk assessment (Table 1), the HCP profile may be significantly affected by the cell line and related process changes; therefore, multiple orthogonal methods were used to evaluate the HCP differences. The total HCP amount was quantified by ELISA for intermediate quality control and lot release (Fig. 6A). The HCPs in the pre- and post-change IBI305 drug substances were both at substantially low levels (below LOQ), and the intermediates of downstream purification steps had lower amounts of HCPs after the change. The pre-change harvest cell culture fluid (HCCF) had averagely 2.6-fold HCP abundance compared with the post-change HCCF, which was consistent with the differences of cell growth states. The results also showed better HCP removal performance of post-change affinity purification process. Both processes lowered HCP amount to several ppm levels after the step of adsorptive depth filters (ADF) and below LOQ by the next CEX purification step. The silver-stained 1D-gel electropherograms showed the same decreasing trend with ELISA (Fig. 6B).

To further assess HCP risk after the cell line change, a bottom-up proteomic analysis was implemented to characterize the HCPs by combining nanoLC-MS/MS with offline high-pH fractionation, which could significantly decrease the wide dynamic range across proteins and improve the sensitivity for HCP identification. Consistent with the ELISA results, a significantly lower number of proteins were detected in the intermediate of each post-change downstream purification step, and the identified proteins showed substantial overlap between the pre- and

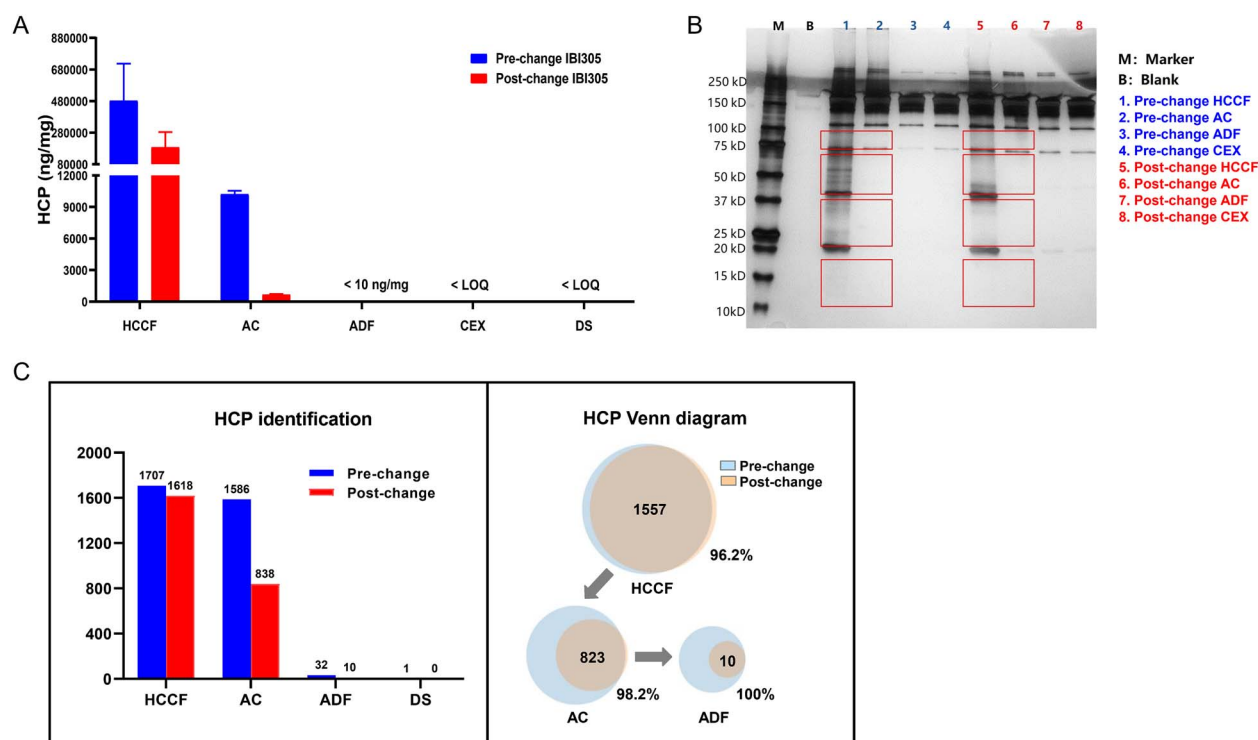


Figure 6. HCP profiling of the pre-change and post-change IBI305 intermediates and drug substances. (A) ELISA assay, (B) silver-stained 1D-gel electropherograms (the protein bands in the borders showed visually different intensities before and after the change) and (C) HCP identification by LC-MS/MS coupled with offline high-pH fractionation.

post-change intermediates (Fig. 6C). After ADF step, only 10 proteins were detected in the post-change intermediates, which were all overlapped with the 32 proteins identified before the change, and no new HCP was identified from the post-change IBI305. Only one protein, the endoplasmic reticulum chaperone BiP, was identified in both pre- and post-change drug substances, and the content was reduced by half after the change.

Biological activities and immunological properties

The biological activity that contributes to the clinical efficacy of bevacizumab is mediated by the neutralization of VEGF-A through the Fab domain. The comparability assessment included potency measured by the VEGF reporter gene bioassay and binding to the target VEGF 165 (as well as VEGF-A) measured by ELISA, as listed in Table 2. The results of VEGF reporter gene bioassay are shown in Fig. 3G. The cell-based bioactivity of the pre-change IBI305 was 90–113%, whereas 88–97% for the post-change IBI305 and 79–123% for Avastin®, all meeting the comparability criteria, and the binding activities of all products met the comparability criteria as well. The binding affinities to Fc receptors (FcRn and FcγRs) and C1q measured by biolayer interferometry (BLI) method demonstrated that both the pre- and post-change IBI305 had similar binding affinity in comparison to Avastin® (Table S4).

Accelerated stability and forced degradation studies

Accelerated stability (25°C) and forced degradation studies under high temperature (40°C) and light exposure were

performed for the comparison of degradation pathways and rates.

The results for general properties of pre- and post-change IBI305 and Avastin® were comparable and within the acceptance criteria. The degradation profiles of all evaluated stability-indicative attributes of IBI305 at 25°C for 6 months were consistent before and after the change. Degradation profiles after incubated at 40°C for 10 days were comparable among pre- and post-change IBI305 and Avastin®. Aggregates and fragments retained unchanged revealed by SEC-HPLC and non-reduced CE-SDS (Table S5). The CEX results showed that the acidic species contents of post-change IBI305 obviously increased by about 9% after 10 days, which was comparable to the pre-change IBI305 but slightly lower than the increasing percentage (11%) of Avastin® (Fig. 7A). The corresponding reductions of the main species for all samples were also observed (Fig. 7B) and minor changes occurred for the basic species (Table S5). This stability difference between IBI305 and Avastin® might be mainly attributed to the different formulation buffers rather than method variation according to our early formulation study that indicated that bevacizumab was more stable in IBI305 formulation buffer compared with Avastin® formulation buffer. The thermal stress showed no impact on biological activities of the pre- and post-change IBI305 and Avastin® (Table S5).

Light exposure at 5000 Lux ± 500 Lux for 10 days induced similar degradation trends among all samples, the results are shown in Table S6. SEC-HPLC results showed that the pre- and post-change IBI305 exhibited a comparable decrease of the main peak (about 5%), and a much faster

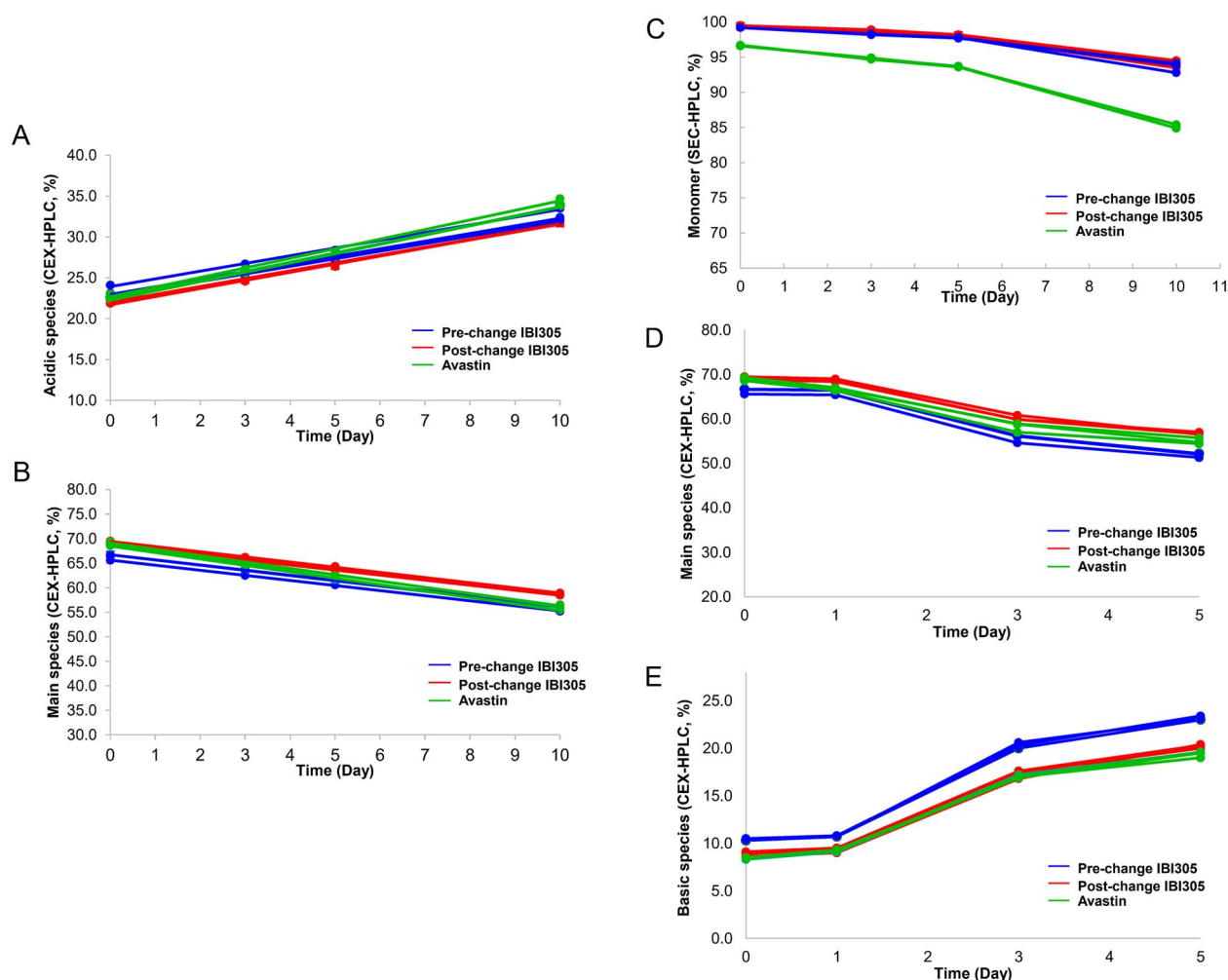


Figure 7. Forced degradation profiles of the pre-change and post-change IBI305 and Avastin[®]. (A, B) Time-course CEX results for high temperature (40°C) and (C–E) time-course SEC and CEX results for light exposure (5000 Lux $\pm$ 500 Lux).

decreasing rate of about 12% in 10 days was observed for Avastin[®] (Fig. 7C), which indicated better light exposure stability for IBI305 than Avastin[®] because of the different formulation buffers. The trends of increase in aggregates and fragments and decrease of the main peak observed by non-reduced CE-SDS were in accordance with SEC-HPLC (Table S6). As for CEX results, the main species of all samples decreased at comparable rates (Fig. 7D), and the basic species increased at similar paces (Fig. 7E). The LC-MS/MS results revealed consistently increased tryptophan and methionine oxidation among all the samples, and corresponding declining trends of binding activities were observed, approaching the lower limit of the quality specification (Table S6). The increased Trp50 and Trp108 oxidation assigned as CQAs mainly contributed to the drops of the binding activities based on the previous structure–function relationship studies (data not shown).

Nonclinical and clinical studies

To support the cell line change, a comprehensive panel of nonclinical comparability studies was conducted, including

three-way comparisons of PD, PK and toxicology profiles of pre- and post-change IBI305 and Avastin[®]. The PD studies included cell-based blocking activity, inhibition of primary HUVEC proliferation and *in vivo* anti-tumor efficacy in HL60 tumor-bearing mice. The PK studies included single-dose PK studies in mice and cynomolgus monkeys, accompanied by immunogenicity testing. The toxicology study included a 4-week repeat-dose toxicity study in cynomolgus monkeys, along with toxicokinetic and immunogenicity tests. Based on the comparable results of toxicological, PK and PD studies (data not shown), it is reasonable to infer that the nonclinical safety and efficacy of pre- and post-change IBI305 and Avastin[®] are all comparable.

Healthy male subjects were selected for the clinical PK bridging of pre- and post-change IBI305 according to the Chinese guidelines for clinical trials of bevacizumab biosimilar and the previous clinical results. A sample size of 50 subjects for each group (100 in total) was based on 85% power to demonstrate the bioequivalence of pre- and post-change IBI305. Since bevacizumab has immunogenicity and a relatively longer half-life, a single-dose study was

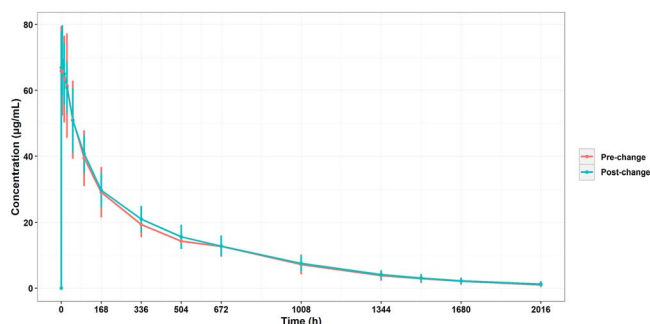


Figure 8. Serum concentration-time profiles (Mean $\pm$ SD) of pre- and post-change IBI305 after single IV dose at 3 mg/kg in healthy subjects.

conducted because of the ethical considerations. In all, 99 subjects received one dose of pre- or post-change IBI305. Sufficient PK data were collected and subjected to comparability assessment from 97 subjects for accurate calculation of $AUC_{0-\infty}$ ($AUC_{0-t}/AUC_{0-\infty} < 80\%$ or $Rsq_adjusted \leq 0.8$). The average $AUC_{0-\infty}$ (CV%) of pre- and post-change IBI305 were 23626.8 (18.1%) h. $\mu\text{g/mL}$ and 24706.7 (16.1%) h. $\mu\text{g/mL}$, respectively. The point estimate and 90% confidence interval for $AUC_{0-\infty}$ were 1.043 (0.986, 1.104) (post-change to pre-change). The serum concentration-time profiles (mean $\pm$ SD) of pre- and post-change IBI305 after single IV dose at 3 mg/kg in healthy subjects were shown in Fig. 8. The sample size estimation assumed a coefficient of variation of 35%. The observed CV% for $AUC_{0-\infty}$ (18.1%) is obviously lower than the assumption. The results indicated the high quality of study and provide extremely sensitivity to evaluate the comparability of pre- and post-change IBI305. During the study period, the incidence of treatment emergent adverse event (TEAE) was comparable in the pre- and post-change IBI305 groups (68.0 vs. 63.3%), and the incidence of TRAE was also comparable (64.0 vs. 63.3%). Meanwhile, no ADA was detected after baseline before and after the change, indicating that pre- and post-change IBI305 had similar immunogenicity.

DISCUSSION

Advances in biomanufacturing processes, especially recombinant protein expression cell lines, drive the development of higher-producing process to improve the drug affordability and availability for biological products. Cell line change might be the most complex process change for bi-therapeutic products. Our work represents the first report of comparability exercise for the post-approval production cell line change. Based on risk assessment, the cell line change and corresponding process adaptations potentially cause significant impacts on a battery of CQAs, particularly on undesired sequence variants, HOS and HCP profile, and therefore bring notable adverse effects to the product quality, safety and efficacy. These high-risk uncertainties incurred the comprehensive comparability study including in-process control, extensive analytical, nonclinical and clinical PK, PD, toxicological and immunogenicity

comparisons. Since IBI305 is a biosimilar, in order to avoid overlooking the potential drift of quality attributes caused by the changes, a three-way comparison of the post-change IBI305 to both the pre-change IBI305 and reference product Avastin[®] was performed, with the similarity acceptance criteria for NDA approval applied as the comparability criteria.

To maximally observe the quality changes and minimize uncertainties, state-of-the-art analytical techniques with high specificity, sensitivity and accuracy were extensively applied and demonstrated the analytical comparability. Lower contents of galactosylation and higher levels of afucosylation were observed after the cell line change. Afucosylation and galactosylation were reported to be positively correlated with antibody-dependent cellular cytotoxicity (ADCC) and complement-dependent cytotoxicity (CDC) of IgG1, respectively. However, because of the fact that bevacizumab only binds to secreted VEGF-A and thus does not exhibit ADCC or CDC [37], the differences of these two glycoforms were suspected to have no adverse impact on the biological functions of IBI305, which was confirmed by the comparable bioactivity.

It has been reported that ¹H-NMR and ¹H-¹³C 2D NMR have much higher sensitivities compared with the typical conformation techniques such as CD and Fourier-transform infrared spectroscopy [38–40], and 2D NMR has been performed to assess the HOS of monoclonal antibodies [41–43]. 1D and 2D NMR techniques were used complementally for HOS analysis of IBI305 and Avastin[®]. To the best of our knowledge, ¹H-¹³C 2D-NMR, with superior ability to address minor conformation change, was the first time applied for HOS assessment in comparability exercise of process change.

Even trace levels of HCPs present in biological product could pose a risk to the safety and efficacy [44–46]. The HCPs in the intermediates and drug substances were particularly monitored and investigated by three orthogonal methods. The qualitative changes of HCP populations influenced by cell line and production process changes were not necessarily detectable by commonly used approaches for measuring total HCP levels, such as ELISA. Mass spectrometry is emerging as a powerful tool for detecting and identifying individual HCP during purification process development [13]. Besides, optimized sample preparation strategies, such as online or offline fractionation, could improve detection sensitivity by peptide fractionation [47, 48]. In this work, advanced nanoLC-MS/MS technique coupled with offline high-pH fractionation, which extensively increases the throughput of HCP identification with low-ppm-level sensitivity, was used for the first time to characterize and compare HCPs in comparability assessment. Only protein BiP, which is widely observed in commercial therapeutic antibodies [49, 50], was observed in the IBI305 drug substance and no new HCP was identified in the post-change product. Estimated by the mass spectra intensities, the abundance of BiP decreased by half after the cell line change. By combining multiple orthogonal methods, the potential risk of HCP for the post-change IBI305 was lowered to minimal level through the in-depth side-by-side

analysis of the pre- and post-change intermediates and drug substances.

Overall, a case of comparability study for post-approval production cell line change was presented. Minor differences in some quality attributes were identified by the analytical comparisons of pre- and post-change IBI305 and Avastin®, whereas no significant impact on product safety and efficacy was expected. Although a comprehensive analytical comparability study was performed to explore the delicate differences existing in the quality attributes, because of the limited method types and detection capabilities, it was still insufficient to ensure that all risks were eliminated. Therefore, the nonclinical and clinical bridging studies were further conducted to minimize the remaining uncertainties. The nonclinical safety and efficacy of pre- and post-change IBI305 and Avastin® were all comparable, and the total evidence of clinical studies in human PK and safety supported the comparability between the pre- and post-change IBI305. Because such a post-approval cell line change was made for the first time in China, multiple rounds of in-depth communications about the scope and extent of the comparability exercise were made between the regulator and manufacturer. Considering the complexity of such post-approval changes, implementation of ICH Q12 guideline and the proposed tools were strongly recommended for faster and more predictable post-approval change management, to streamline the procedure and approval of post-approval changes.

Along with the rapid development of biological product, more and more manufacturers would undergo higher-titer cell line change after approval to expand the affordability and reduce the manufacturing cost. The work presented here provides manufacturers a good case as reference to perform comparability study for post-approval cell line change of biological products, especially for biosimilars.

SUPPLEMENTARY DATA

[Supplementary Data](#) are available at ABT Online.

CONFLICT OF INTEREST STATEMENT

The authors ZW, WH, KW are employees of National Medical Products Administration, GX, CY, LW are employees of National Institute for Food and Drug Control, WH, SZ, DK, MHX, XC are employees of Innovent Biologics, Inc., and declare that they have no other conflicts of interest that might be relevant to the contents of this article.

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All studies were conducted at Innovent Biologics and supported by Innovent Biologics. Regulatory reviews at CDE of NMPA and NIFDC were performed independently.

DATA AVAILABILITY

The data from this study are available from the corresponding author upon reasonable request.

AUTHOR CONTRIBUTIONS

Zhouyi Wu (Data curation-Equal, Investigation-Equal, Writing—original draft-Equal, Writing—review & editing-Equal), Gangling Xu (Data curation-Equal, Formal analysis-Equal, Writing—original draft-Equal, Writing—review & editing-Equal), Wu He (Conceptualization-Equal, Formal analysis-Equal), Chuanfei Yu (Data curation-Supporting, Formal analysis-Supporting, Investigation-Supporting), Wanqiu Huang (Data curation-Equal, Formal analysis-Equal, Investigation-Equal, Methodology-Equal, Writing—original draft-Equal), Shirui Zheng (Data curation-Equal, Formal analysis-Equal), Dian Kang (Data curation-Equal, Formal analysis-Equal), Michael H. Xie (Conceptualization-Equal, Data curation-Equal, Formal analysis-Equal, Writing—original draft-Equal, Writing—review & editing-Equal), Xingjun Cao (Conceptualization-Lead, Data curation-Equal, Formal analysis-Equal, Methodology-Equal, Writing—original draft-Lead, Writing—review & editing-Lead), Lan Wang (Conceptualization-Lead, Investigation-Lead, Writing—review & editing-Lead), and Kaikun Wei (Conceptualization-Lead, Writing—review & editing-Lead)

ETHICS AND CONSENT

The clinical trial was registered in www.chinadrugtrials.org.cn, CTR20211576, and approved by Ethics Committee of the First Affiliated Hospital of Soochow University. Before undergoing study specific procedures, all subjects signed informed consent form.

ANIMAL RESEARCH

All mice experiments were performed in accordance with regulations for the use of laboratory animals at Innovent Biologics and were approved by the Institutional Animal Care and Use Committee (IACUC-01). All monkey experiments were approved by IACUC and performed according to the regulation of AAALAC.

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